ADDITIONAL PROJECT

Single-Axis Solar Tracking System

Light-Differential Servo Control for Solar-Panel Alignment

CONTEXT	ROLE	APPLICATION	EVIDENCE
Embedded Control Prototype	**Embedded Systems Developer**	**Panel Orientation**	**Firmware + Prototype Video**

Project Overview

Implemented a compact feedback controller that aligns a solar panel along one mechanical axis using the light-intensity difference between two LDR sensors. An Arduino-compatible controller samples both analog channels, applies a deadband to reject small imbalances, and advances a servo toward the brighter side while enforcing software travel limits. The design demonstrates the full embedded loop from sensing and decision logic to constrained electromechanical actuation, with serial diagnostics provided for calibration and functional observation.

Single-axis tracking prototype during a light-direction demonstration.

Engineering implementation

- Integrated two analog LDR channels on `A0` and `A1` to estimate the direction of higher incident light.
- Developed a deterministic comparison routine with a 40-count tolerance band to suppress unnecessary servo movement.
- Commanded one-degree corrections through the Arduino Servo library, starting from a centred 90-degree position.
- Constrained the commanded travel to 15–165 degrees to protect the intended mechanical range.
- Added 9600-baud serial reporting of both sensor readings and the commanded servo angle for calibration and debugging.

CORE RESPONSIBILITY

Transforming differential light measurements into bounded, incremental servo commands for panel alignment.

IMPLEMENTED BEHAVIOR — EVIDENCE VERIFIED

The firmware establishes the single-axis control logic, while the portfolio video shows the physical prototype being exercised with a handheld light source. No calibrated tracking-accuracy or energy-gain data was supplied.

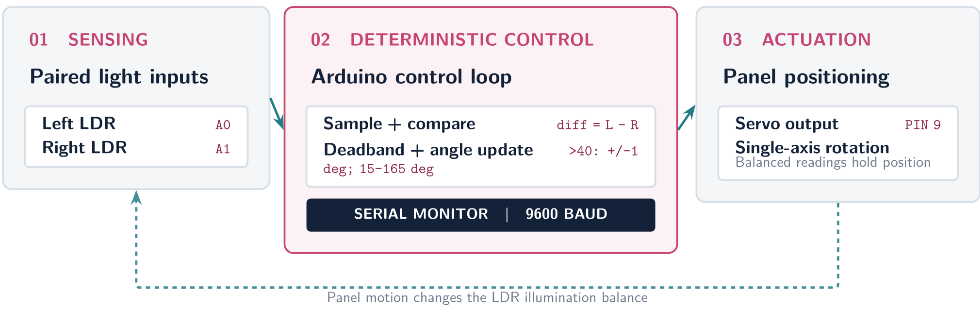
SYSTEM AT A GLANCE

Sense Two analog LDR inputs

Decide Difference, tolerance, direction

Actuate One servo on digital pin 9

Control Architecture



Implemented control loop from paired LDR sensing to constrained servo actuation and serial diagnostics.

PROPOSED 2026 ARCHITECTURE EVOLUTION — NOT YET IMPLEMENTED

- Add a second axis with position feedback, homing switches, and motor-current protection for repeatable, mechanically safe motion.
- Instrument panel voltage and current and integrate an MPPT charging stage to validate harvested power against a fixed-panel baseline.
- Combine astronomical sun-position calculation with LDR trim so deterministic tracking remains stable during cloud cover and uneven shading.
- Add an ESP32-class supervisory layer for local logging, secure telemetry, and configuration while retaining real-time control on the microcontroller.

TECHNOLOGIES

Arduino-Compatible MCU · Embedded C/C++ · Analog ADC · LDR Sensing · Servo Control · Deadband Logic · Serial Diagnostics